

LAMAR, CO, USA

WMO: 724636

Lat: 38.070N

Lon: 102.688W

Elev: 1129

StdP: 88.48

Time Zone: -7.00 (NAM)

Period: 95-19

WBAN: 03013

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.4	-14.1	-21.2	0.6	-12.9	-18.4	0.8	-8.0	12.8	5.6	11.2	6.3	2.9	210	0.625

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.8	38.1	18.6	37.0	18.7	35.3	18.5	21.6	31.3	20.9	30.5	20.4	30.1	5.5	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.0	15.9	23.7	18.3	15.1	23.4	17.7	14.5	23.1	68.2	30.7	65.8	30.7	63.5	30.0	26.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.9	11.3	9.6	DB	-22.8	40.9	4.2	1.5	-25.8	42.0	-28.3	42.8	-30.6	43.7	-33.6	44.7	(4)
(5)			WB	-23.1	23.1	4.1	1.1	-26.1	23.9	-28.5	24.5	-30.8	25.1	-33.7	26.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.2	-0.9	1.2	6.6	11.4	17.0	23.2	26.1	24.5	20.2	12.1	5.0	-0.7	(6)	
	DBStd	10.69	5.72	6.08	5.45	4.96	4.62	3.88	2.76	2.83	4.34	5.04	5.08	5.60	(7)	
	HDD10.0	1294	337	249	132	42	5	0	0	0	1	35	160	333	(8)	
	HDD18.3	2956	595	480	366	211	79	6	1	1	30	198	399	591	(9)	
	CDD10.0	2097	0	3	25	84	222	395	500	450	306	101	11	1	(10)	
	CDD18.3	718	0	0	1	3	38	151	242	192	85	6	0	0	(11)	
	CDH23.3	10354	0	5	63	231	808	2159	3202	2322	1276	271	18	0	(12)	
	CDH26.7	5516	0	1	11	67	359	1194	1894	1262	633	93	1	0	(13)	
(14) Wind	WSAvg	4.2	3.5	4.0	4.5	5.3	4.9	4.8	4.4	4.0	4.1	3.9	3.6	3.6	(14)	
(15) Precipitation	PrecAvg	392	11	11	23	33	61	60	59	56	32	22	15	9	(15)	
	PrecMax	581	36	53	119	128	177	168	200	191	100	121	50	44	(16)	
	PrecMin	244	0	0	0	0	0	5	8	2	1	0	0	0	(17)	
	PrecStd	86	10	12	21	30	43	40	39	44	25	24	15	9	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.6	23.4	28.0	31.3	35.9	40.7	40.0	38.9	37.4	32.2	25.7	19.7	(19)	
		MCWB	7.4	8.9	11.2	12.9	15.9	17.1	19.2	18.5	18.0	15.0	11.0	8.2	(20)	
	2%	DB	16.3	19.5	25.1	28.6	33.0	37.7	38.4	37.0	35.0	29.3	22.4	15.9	(21)	
		MCWB	6.0	7.5	10.0	12.0	15.4	17.5	19.0	19.1	17.3	13.8	9.7	6.0	(22)	
	5%	DB	12.7	16.2	22.1	26.2	30.8	35.7	37.2	35.2	32.8	26.7	19.0	12.5	(23)	
		MCWB	4.5	6.0	9.3	11.5	15.2	17.6	19.3	19.0	17.1	13.3	8.5	4.6	(24)	
	10%	DB	9.0	12.4	18.4	23.0	28.0	33.2	35.8	33.2	30.9	23.0	15.6	9.0	(25)	
		MCWB	2.9	4.3	8.0	10.8	14.6	17.5	19.5	19.2	16.9	12.0	6.9	3.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.1	9.6	12.8	15.1	19.0	21.7	22.5	22.4	20.6	17.1	12.3	8.6	(27)	
		MCDB	17.9	21.5	24.0	24.4	29.4	30.8	32.4	31.3	29.8	26.5	21.7	18.7	(28)	
	2%	WB	6.4	7.9	11.1	13.7	17.7	20.5	21.7	21.6	19.5	15.5	10.4	6.5	(29)	
		MCDB	15.8	18.5	22.0	24.2	27.5	30.1	31.7	30.9	29.2	24.7	20.2	15.1	(30)	
	5%	WB	4.8	6.4	9.8	12.5	16.7	19.7	21.0	20.8	18.7	14.1	9.0	4.9	(31)	
		MCDB	12.3	15.7	20.4	22.8	26.7	29.5	31.2	29.9	28.9	23.7	18.5	12.1	(32)	
	10%	WB	3.1	4.7	8.4	11.4	15.7	18.9	20.5	20.2	17.8	12.9	7.4	3.2	(33)	
		MCDB	8.4	11.8	17.6	21.4	25.3	28.9	30.7	29.4	27.4	22.0	15.2	8.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	16.7	16.9	18.4	17.5	17.5	17.4	16.8	16.1	17.4	18.4	18.1	16.7	(35)	
		MCDBR	22.6	23.8	24.4	23.6	21.6	20.7	19.7	19.0	20.9	23.7	24.2	22.0	(36)	
	5% WB	MCWBR	13.6	13.2	11.8	10.3	8.4	6.5	5.5	5.5	7.2	10.6	12.7	13.4	(37)	
		MCWBR	21.3	22.6	22.4	20.1	18.8	17.6	16.4	16.2	17.9	20.1	22.7	21.2	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.249	0.258	0.285	0.315	0.333	0.350	0.373	0.360	0.325	0.285	0.261	0.250	(40)	
		taud	2.583	2.574	2.496	2.423	2.435	2.416	2.404	2.405	2.489	2.580	2.569	2.576	(41)	
	Edn at Noon		961	990	984	964	946	925	900	907	926	943	935	936	(42)	
		Edh at Noon	71	82	98	113	114	116	117	114	98	81	71	67	(43)	
(44) All-Sky Solar Radiation	RadAvg		2.73	3.56	4.92	6.06	6.99	7.61	7.50	6.63	5.52	4.12	3.03	2.40	(44)	
	RadStd		0.20	0.33	0.39	0.32	0.38	0.32	0.25	0.27	0.27	0.34	0.22	0.15	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (18 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page